

Development of an Intelligent Predictive Model for Managing the General Warehouse of a Dairy Industry Using Neural Networks

- Jeniffer Mishelle Arce Campoverde
- Kevin Fernando Fajardo Parra
- Thesis Advisor: Pablo Andrés Flores Sigüenza

Abstract

A neural-network-based predictive model was developed to support procurement planning for critical supplies in a dairy industry warehouse, using historical data (Jan 2024–Sep 2025). The methodology included a systematic review and diagnosis, model development and comparison, and scenario-based validation. MLP, LSTM, and a hybrid MLP+LSTM were evaluated with Macro-F1 and MAE_cond, showing stable performance for intermittent demand as a decision-support tool.

Objectives

General Objective

To develop a predictive model of procurement needs based on machine learning and neural networks to optimize general warehouse management in a dairy industry.

Specific Objective

To evaluate performance under different scenarios and metrics in order to validate the robustness of the approach.

To analyze the current state of general warehouse management and identify procurement and consumption issues.



To design, train, and compare architectures (two-stage linear regression, MLP, LSTM, and the hybrid MLP+LSTM) to forecast demand.

Specific Objective

Specific Objective

Methodology

1

Phase 1. Systematic review and initial diagnosis
Literature review to identify variables/models and diagnosis of historical data to define critical inputs.

2

Phase 2. Predictive Model Construction
Data preparation and model training (comparison and selection of the final approach).

3

Phase 3. Validation and scenario analysis
Testing the model in alternative scenarios to assess stability and support recommendations.

Curricular Integration

Analytical Statistics and Design of Experiments (data cleaning and analysis), Logistics and Supply Chain Management (procurement and turnover criteria), and Machine Learning (model design and validation).

Applied Engineering

ISO 9001 (traceability, document control and changes) + ISO/IEC 27001 (data confidentiality) + ISO 31000 and ISO/IEC 23894 (project and AI risk management) + ISO 22000 (traceability/safety as a context for the dairy sector).

The design considered safety and well-being (confidentiality and responsible use of data), social/operational impact (better planning and fewer reactive decisions) and economic factor (reduction of costs due to overstock and shortages without immediate investment).

Keywords

machine learning

neural networks

inventory management

modelo predictivo

dairy industry

Results

Historical records from January 2024 to September 2025 were used, focusing on critical supplies based on their operational impact and replenishment lead time.

The hybrid MLP+LSTM approach was selected for its better balance: MLP supports occurrence detection, while LSTM supports magnitude estimation.

In scenario analysis, magnitude performance improved consistently under the −5% case and with P80 smoothing: in both cases, MAE_cond on the test set decreased for 9 out of 10 items, and the average error dropped by about 10% compared to the baseline (P85).

Conclusions

Conclusion 1

A predictive model (machine learning + neural networks) was developed to forecast the consumption of critical supplies in a real-world context with high variability and intermittency.

Conclusion 2

The diagnosis revealed overstocking and stockouts due to misalignment Sales, Production, and Purchasing/Warehouse use, as well as the difficulty of anticipating future needs under irregular demand patterns.

Conclusion 3

When comparing alternatives (RL, MLP, LSTM), some performed better at occurrence detection while others performed better at magnitude estimation; therefore, the hybrid MLP+LSTM approach was selected as the best overall balance.

Conclusion 4

In scenario analysis, the −5% and P80 cases improved magnitude performance (test MAE_cond) for 9 out of 10 items and reduced the average error by about 10%; the multivariable-by-presentation scenario worsened performance, so it is not considered stable. This supports using smoothed historical data (P85/P80) and complementary policies (e.g., safety stock).

Contacts

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References

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